A 40nm 24.85-to-80.59 TFLOPS/W FP8 Computing-in-Memory Macro with Outlier-Aware Hierarchical-Alignment for Edge Transformers

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Abstract

This paper presents a digital Compute-in-Memory (CIM)-based macro tailored for edge Transformer acceleration. It has three features: 1) A Hierarchical Dual-Exponent Alignment scheme that separates outliers from normal values, achieving near-ideal accuracy with minimal overhead. 2) A Sign-Magnitude Input Serial Pipelined Aligner (SMSPA) reduces area and power by 41.1% and 34.1%. 3) A Shift-Concat based Sign-Inverse Adder Tree (SSAT) saves accumulation area and power by 25.9% and 18.2%. The macro achieves 24.85-80.59 TFLOPS/W in FP8. Measurements demonstrate up to 6.66× improvements in energy efficiency, compared to state-of-the-art post-alignment FP-CIM schemes, while maintaining high accuracy on different tasks.

Introduction

Transformers dominate edge applications, yet the significant outlier distributions in their activations and weights render conventional INT quantization prone to severe clipping or rounding errors (Fig. 1). Floating-Point CIM (FP-CIM) emerges as a promising solution offering high dynamic range [1-11], but existing implementations face a fundamental accuracy-efficiency trade-off (Fig. 2). Pre-alignment schemes enable the use of compact INT-MACs but suffer from accuracy loss due to shared exponents. Conversely, post-alignment schemes preserve accuracy but incur prohibitive area and energy penalties due to massive shifting and wide accumulation logic. Bridging this gap to achieve post-alignment accuracy with pre-alignment efficiency faces two primary hardware challenges: (C1) The discrepancy between parallel alignment and bit-serial MAC operations in pre-alignment schemes creates inevitable pipeline stalls, severely degrading utilization [1-3]. (C2) The redundant addition in post-alignment schemes consumes excessive overhead due to sign extension and invalid switching activities [4-6].

To address these issues, this paper introduces an outlier-aware FP8 CIM macro with the following key features: 1) A Hierarchical Dual-Exponent Alignment (HDEA) strategy that separates outliers to achieve near-ideal accuracy with minimal alignment overhead. 2) A Sign Magnitude Input Serial Pipelined Aligner (SMSPA) that leverages serial pipelining to eliminate stalls. 3) A Shift-Concat based Sign-Inverse Adder Tree (SSAT) that removes redundant signed addition to minimize hardware cost.

Proposed Overall Architecture

Fig. 3 illustrates the overall architecture of the proposed CIM macro, consisting of an SMSPA, 16 CIM subarrays, and 16 SSAT. The CIM subarray utilizes standard 6T SRAM bitcells configured for 64×4b. To address the precision loss in pre-alignment schemes caused by outliers, the **Hierarchical Dual-Exponent Alignment (HDEA)** strategy (Fig. 4) is introduced. Instead of globally aligning to E_{max}, HDEA dynamically classifies inputs: outliers align to E_{max} (Flag F=1), while normal values align to E_{mid}= E_{max}-ΔE (F=0). This separation limits the left-shift amount to $((F_A+F_W)\times\Delta E)$ for the adder tree, significantly limiting bit growth. Consequently, the width of the adder tree is reduced to $B_W + 2\times\Delta E$ (e.g., 8b), effectively eliminating the area penalty of the 19b adders required in the post-alignment scheme. Finally, we explored the design space of ΔE to balance precision and area. The simulation results confirm ΔE =2 as the sweet spot for achieving accuracy with minimal overhead.

Fig. 5&6 detail the **Sign Magnitude Input Serial Pipelined Aligner (SMSPA)**, designed to resolve the timing mismatch between parallel alignment and bit-serial MAC. Conventional pre-alignment designs incur severe pipeline stalls due to the latency of exponent difference calculation, mantissa shifting, and the 2's complement (2's C) conversion. To eliminate these overheads, SMSPA introduces a fully serialized pipeline. First, the serial E_{max} finder employs a masked unit to compute the global maximum on-the-fly. Exponent bit E_i that is inconsistent with the current E_{max} bit is dynamically masked into the or tree in subsequent bits. Second, the difference accumulator integrates a saturation check against a threshold T. If the difference (E_{diff}) exceeds the limit, the value saturates, and the subsequent shifting operations are gated to eliminate redundant switching power. Finally, leveraging the native SM format allows replacing area-intensive spatial barrel shifters with compact temporal shifters, while reducing the extra sign input cycles. The logical right-shift is efficiently implemented as a time-domain delay using a DFF chain. The SMSPA reduces latency, area, and energy by 20%, 41.1%, and 34.1%, respectively, compared to the parallel baseline.

Fig. 7& 8 present the **Shift-Concat based Sign-Inverse Adder Tree (SSAT)**, addressing the hardware inefficiency of post-alignment addition. Conventional signed adder trees suffer from excessive switching activity and bit-width expansion due to redundant sign extensions. First, to seamlessly support the SM inputs of SMSPA, an activation sign handling unit is developed to perform the negation operation on the partial product (PP) when the activation sign is one. Second, the first addition stage employs a shift-concat adder. By monitoring the Shift Difference (SD), the operation is arithmetically decomposed: overlapping high 4-bit are added, while non-overlapping low 4-bit are directly concatenated. This topology replaces area-hungry global barrel shifters with lightweight local logic, eliminating invalid switching activity. Finally, for subsequent stages, the SSAT replaces conventional signed adders with sign-inversion (SI) unsigned adders. Specifically, the sign bits of partial sums are inverted in the second stage to convert signed operations into unsigned addition, effectively decoupling sign processing from magnitude summation. The correct sign is inverted to restore at the final addition stage. Consequently, the SSAT reduces the area by 25.9% and energy by 18.2% compared to baseline.

Implementation Results and Conclusions

Fig. 9 presents the evaluation results. The improvement breakdown shows the proposed SMSPA and SSAT features cumulatively boost area and energy efficiency by 1.50× and 1.71× over the baseline. Compared to conventional post-alignment schemes, this work achieves remarkable 5.34 × and 4.78 × gains in energy and area efficiency, respectively, with negligible overhead versus pre-alignment. In terms of accuracy, the design significantly outperforms pre-alignment schemes in different tasks, while approaching the post-alignment schemes. Fig. 10 shows that the fabricated macro in 40nm can operate at a 0.8-1.1V supply and 100-326 MHz. The peak energy efficiency is 80.59 TFLOPS/W. Thanks to the proposed hierarchical alignment and efficient adder tree, our design demonstrates 2.81× to 6.66× higher energy efficiency than state-of-the-art post-alignment FP-CIM macros. [4, 5].

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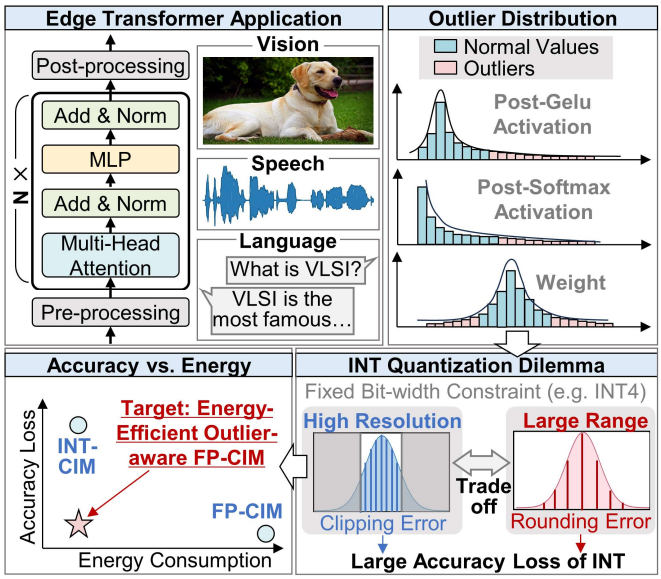


Fig. 1. Transformer models have outlier distributions, leading to severe accuracy loss in the INT quantization.

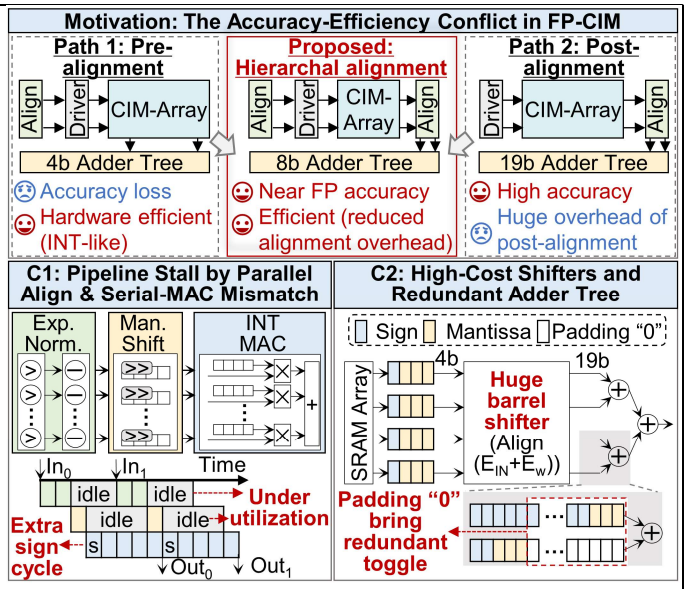


Fig. 2. The accuracy-efficiency conflict motivation and two critical challenges of existing FP-CIM.

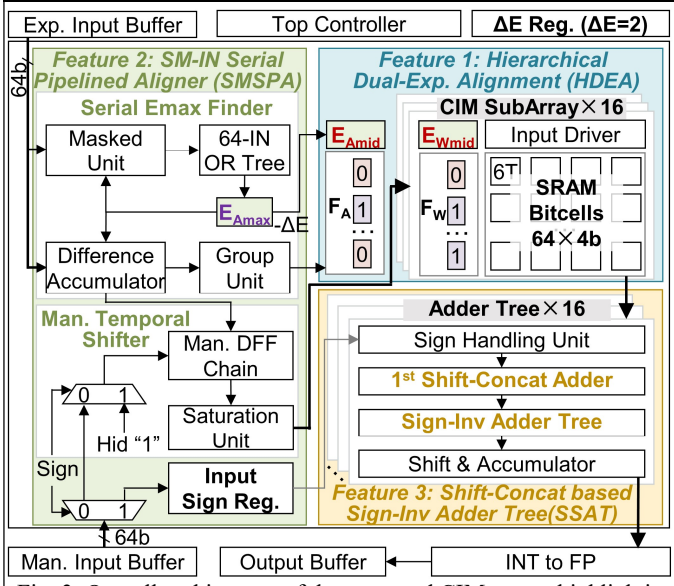


Fig. 3. Overall architecture of the proposed CIM macro highlighting the SMSPA, HDEA, and SSAT features.

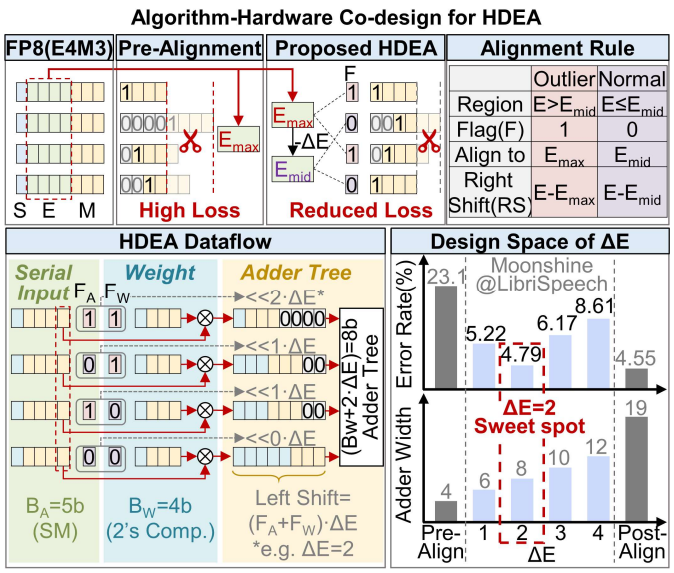


Fig. 4. Proposed Hierarchical Dual-Exponent Alignment (HDEA) strategy and dataflow for reducing precision loss.

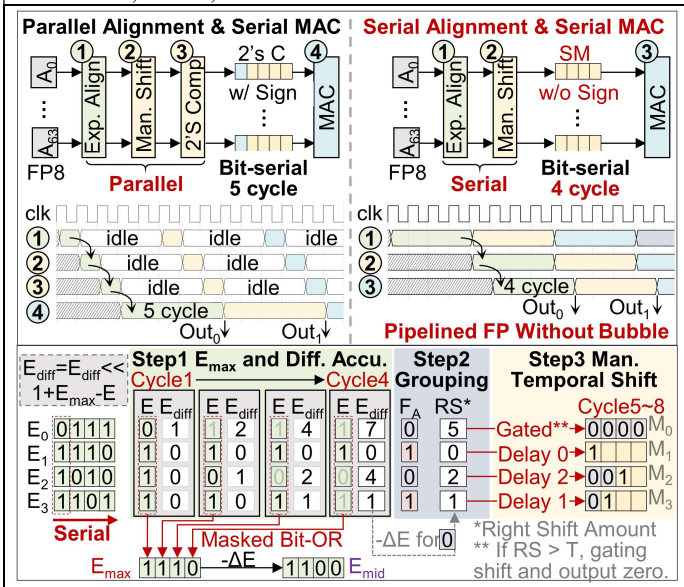


Fig. 5. Schematic of the Sign Magnitude Input Serial Pipelined Aligner (SMSPA), mitigating pipeline bubbles and extra sign cycles.

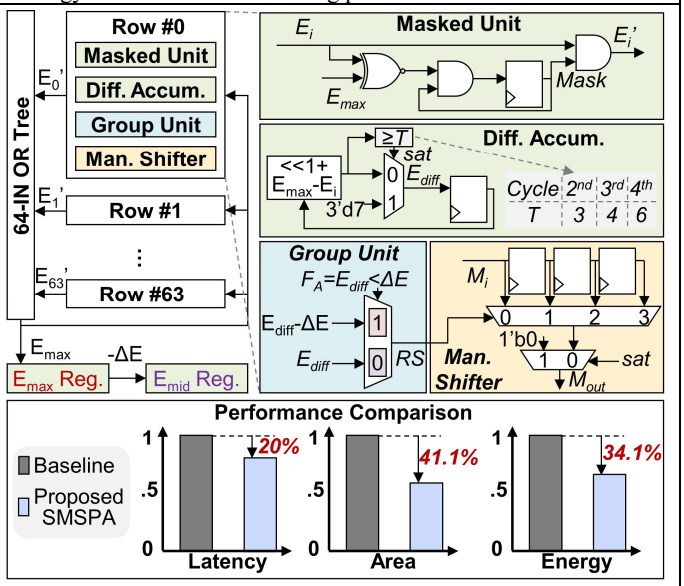


Fig. 6. Proposed SMSPA circuit implementation details and performance comparison.

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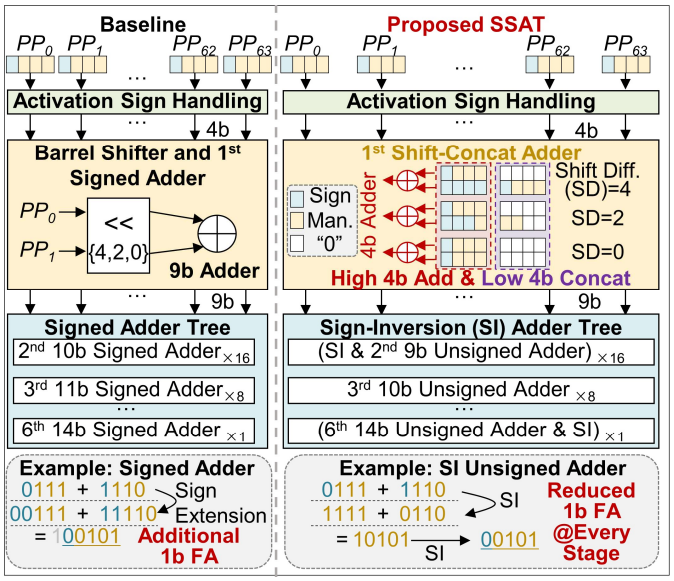


Fig. 7. Shift-Concat based Sign-Inv Adder Tree (SSAT) method that reduces redundant adder and sign extension.

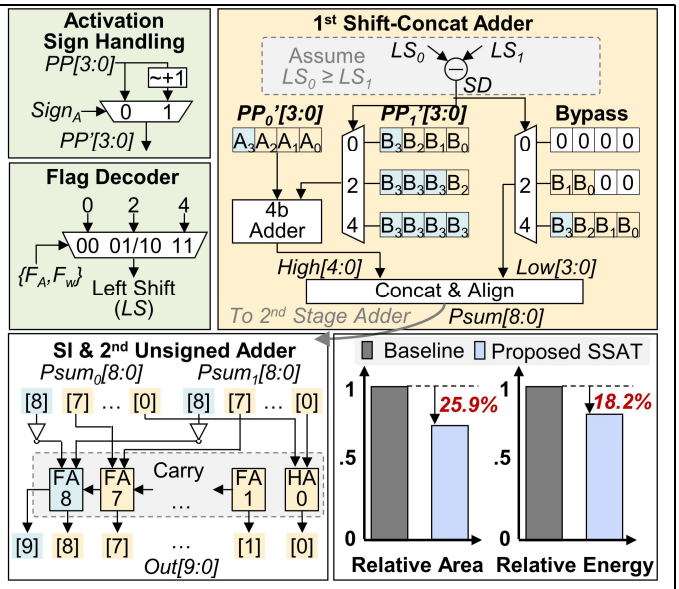
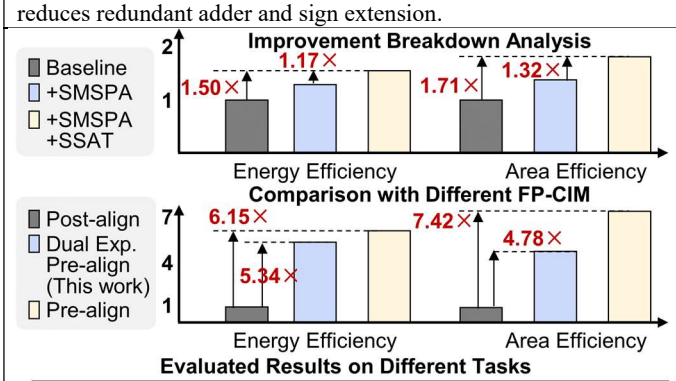


Fig. 8. Detailed proposed SSAT circuit implementation and experimental results comparison.

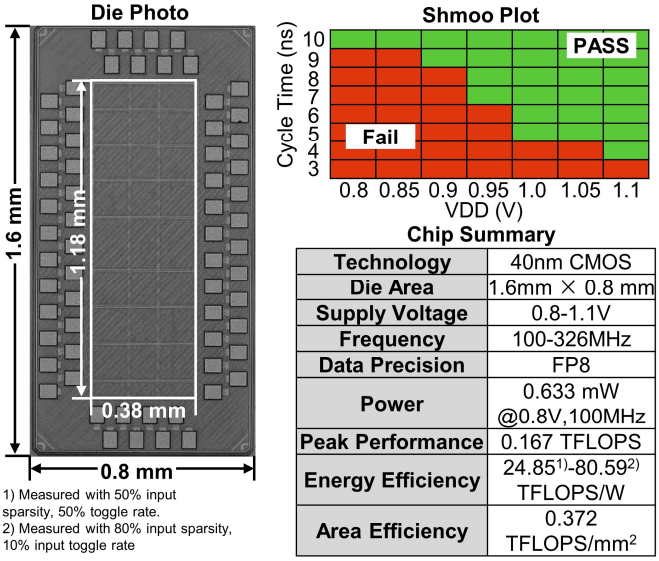


Evaluated Results on Different Tasks

Model		Moonshine-base	DeiT-base	SmolLM2-135M
Parameter		61M	86M	135M
Dataset		LibriSpeech	ImageNet-1K	ARC-easy
Metrics[1)]		WER (%)	Accuracy (%)	Accuracy (%)
FP8	Post-align	4.55	81.67	55.81
	Pre-align	23.07	80.41	45.66
	This work	4.79	81.34	54.88

1) Lower is better for WER (Word error rate), and higher is better for accuracy.

Fig. 9. Performance improvement breakdown and comparison with different alignment FP-CIMs, and accuracy results of three datasets.



1) Measured with 50% input sparsity, 50% toggle rate.
2) Measured with 80% input sparsity, 10% input toggle rate

Chip Summary

Technology	40nm CMOS
Die Area	1.6mm × 0.8 mm
Supply Voltage	0.8-1.1V
Frequency	100-326MHz
Data Precision	FP8
Power	0.633 mW @0.8V,100MHz
Peak Performance	0.167 TFLOPS
Energy Efficiency	24.85[1)]-80.59[2)] TFLOPS/W
Area Efficiency	0.372 TFLOPS/mm²

Fig. 10. Die photo, Shmoo plot and chip summary table.

TABLE I. Comparison Table

	ASSCC' 24[1]	ISSCC' 25[2]	ISSCC' 24[3]	ESSCIRC' 23[4]	VLSI' 25[5]	This Work
Technology	28nm	28nm	22nm	28nm	3nm	40nm
FP Precision	BF16	BF16	FP16, BF16	FP8, BF8	FP16	FP8
CIM Type	Digital	Digital	Analog	Digital	Digital	Digital
CIM Memory	128Kb	64Kb	16Mb	4Kb	153Kb	4Kb
Area [mm²]	0.212	0.136	8.2	1.56	0.042	1.28
Supply Voltage [V]	0.6-0.9	0.55-0.9	0.7-0.8	0.55-1.2	0.41-1.1	0.8-1.1
Exponent Alignment	Pre-align	Pre-align	Pre-align	Post-align	Post-align	Dual-exp Pre-align
Support Transformer	No	Yes	No	No	Yes	Yes
Energy Efficiency [TFLOPS/W]	19.8-45.3	17.83-62.84	31.2-65.5 (BF16)	N/A-12.1 (FP8)	N/A-28.6	24.85-80.59
Area Efficiency [TFLOPS/mm²]	2.015	N/A	0.104	0.427	17.4	0.372

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